

Nikolai Madlener

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EDUCATION

Technical University of Munich

Master of Science in Informatics (Computer Science)

- Focus on Software Engineering, Distributed Systems, and Machine Learning

October 2021 – September 2024

Munich, Germany

Stanford University

Master's Thesis and Visiting Researcher

- Design and implementation of a semi-automated medication recommendation system for hypertension patients

March 2024 – September 2024

Stanford, USA

Peking University

General Visiting Student (Semester abroad)

- Co-organized and spoke at Peking University's first Exchange Cafe about LLMs and ChatGPT

February 2023 – July 2023

Beijing, China

Technical University of Munich

Bachelor of Science in Informatics (Computer Science)

October 2017 – June 2021

Munich, Germany

EXPERIENCE & INTERNSHIPS

Stanford Mussallem Center for Biodesign

Software Engineer (Full-Time)

- Full stack and iOS development for the StanfordSpezi open-source framework using Typescript, Swift and FHIR
- Lead custom development of iOS health app for our partner Johnson & Johnson using SwiftUI and ResearchKit

September 2024 – Present

Stanford, USA / Remote

Weavs GmbH

Software Engineer (Working Student)

- Full stack development of several features of Citymonitor, a smart-city SaaS platform using Elixir and Phoenix

October 2023 – March 2024

Dornbirn, Austria

Technical University of Munich, Digital Health Group

Software Engineer (Academic Assistant)

- Mobile development of an iOS app that improves users' fitness and health by supporting multimodal navigation
- Refactored existing app architecture and improved average route search performance by over 50%

October 2021 – March 2023

Munich, Germany

ALEAS AG

Software Engineer (Working Student)

- Full stack development with .NET for internal and customer-facing software with 2500+ active users
- Mobile development of a cross-platform team-challenge app with React Native and the Firebase platform

July 2019 – November 2022

Ruggel, Liechtenstein / Hybrid

Tesla, Inc.

Logistics Software System Configuration (Internship)

- Initial configuration of the material flow software system for intralogistics at the Tesla Giga Berlin factory
- Requirements engineering and architectural conception based on exchange with material flow planners

March 2021 – May 2021

Grünheide, Germany

AWARDS & EXTRACURRICULAR EXPERIENCE

hackaTUM 2022, 2023 & 2024 Winner

- Overall winner of hackaTUM 2022 & 2023, one of EU's biggest hackathons, out of 220 teams (800+ participants)
- 1st place in the 'City of Munich' 2023 & 2024 challenge and 2nd place in the 'Sixt' 2022 challenge

November 2022, 2023, 2024

MINGA Mentor

- Mentor for incoming exchange students to guide them during their time in Munich and at TUM

August 2022 – February 2023

Ferienakademie 2022 – "Decentralized Decision Making"

- Research summer school in cooperation with Siemens for selected students aimed at exploring innovative ideas
- Development of a self-organized robot delivery system with focus on decentralized decision making using MQTT

September 2022

SKILLS

Languages: German, English (C1), Chinese (HSK1)

Programming Languages: Swift, TypeScript, Elixir

Frameworks & Tools: SwiftUI, React, Phoenix, PostgreSQL, Firebase, Docker, Git